

DELFT UNIVERSITY OF TECHNOLOGY

ANALOGUE CMOS DESIGN 1
EE4520

Assignment 2

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This document presents the design of a discrete-time fully differential amplifier, respecting certain constraints and specifications.

Table of results

Item	Design item	Unit	Specs	Achieved
1	Design number	33		
2	SNR	[dB]	68.895	68.955
3	A_{settle}	[dB]	60	60
4	T_{settle}	[ns]	580	579.5
5	BW_{cl} from open-loop AC simulations	[MHz]	—	1.64
6	BW_{cl} from closed-loop AC simulations	[MHz]	—	1.59
7	Time-constant τ_{cl} from closed-loop AC sims	[ns]	—	100
8	T_{30dB} (time to reach an accuracy of 30dB)	[ns]	—	213
9	T_{50dB} (time to reach an accuracy of 50dB)	[ns]	—	450
10	Time-constant τ_{cl} from transient sim	[ns]	—	102.9
11	Power Dissipation	[mW]	low	0.0227
12	$I(V_{dd})$	[mA]	low	0.0126
13	Total Integrated Noise	[uV]	—	302.62
14	Input Step Voltage	[mV]	200	200
15	Output Step Voltage	[V]	1.2	1.2
16	C_{in}	[pF]	—	0.849
17	C_{fb}	[pF]	—	0.142
18	C_{load}	[pF]	—	0.849
19	C_{CM}	[pF]	—	0.142
20	R_{big}	[GΩ]	big	100
21	C_{big}	[nF]	big	1000
22	I_{b1}	[mA]	—	—
23	I_{b2}	[mA]	—	—
24	Bonus points on FoM_dB	[dB]	0.066	0.066
25	$FoM_{lin} = 6.28 \cdot Power \cdot T_{8.6dB} / (SNR(lin))^2$	[J]	—	$1.87E-18$
26	$FoM_{dB} = -10\log_{10}(FoM_{lin})$	[dB]	—	177.34942

Schematic with all node voltages

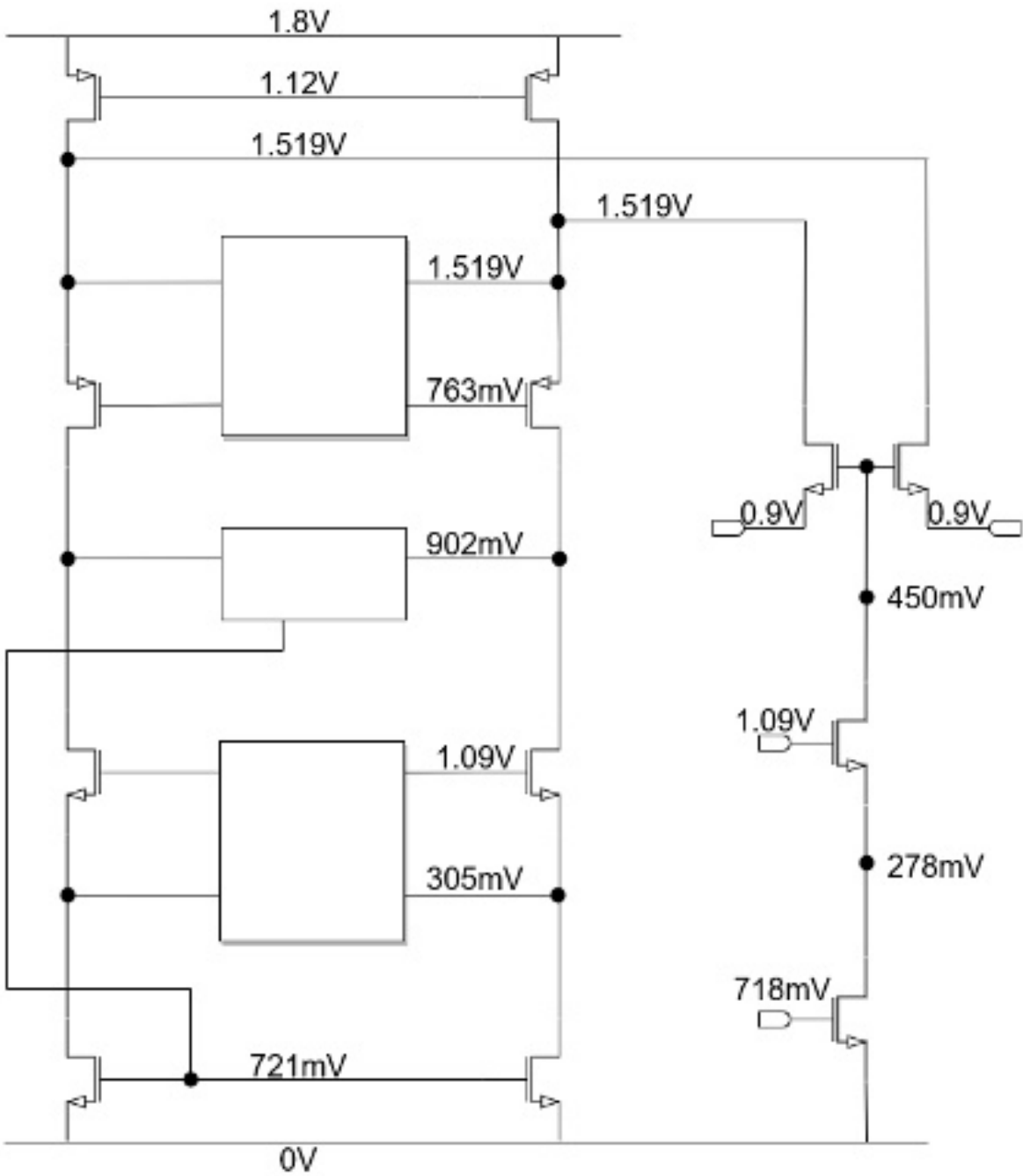


Figure 1: Voltages for each node

Schematic with all branch currents

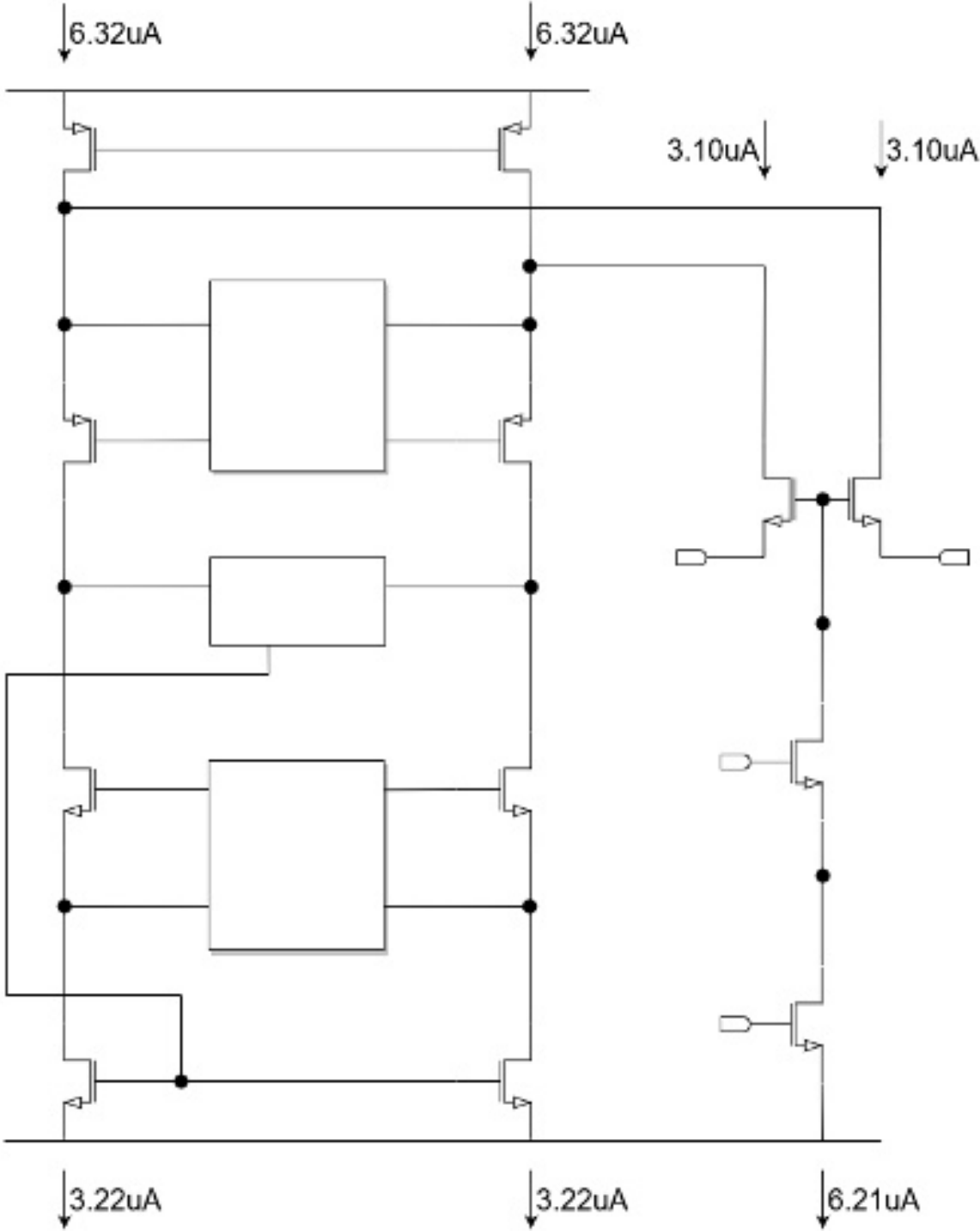


Figure 2: Currents for each branch

Bode-Diagram of A and $A\beta$

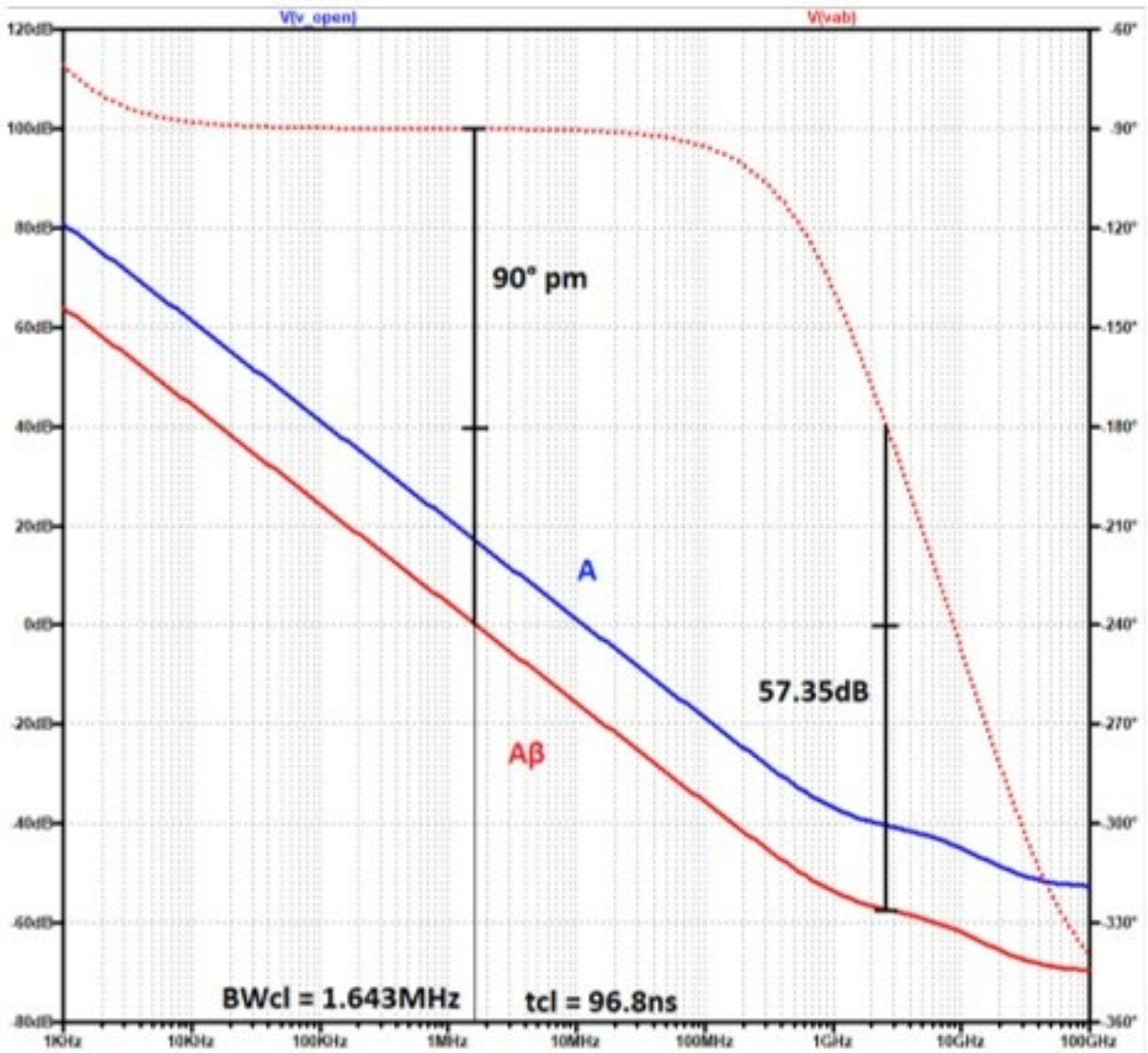


Figure 3: AC response of Open-loop

Bode-Diagram of closed-loop Gain

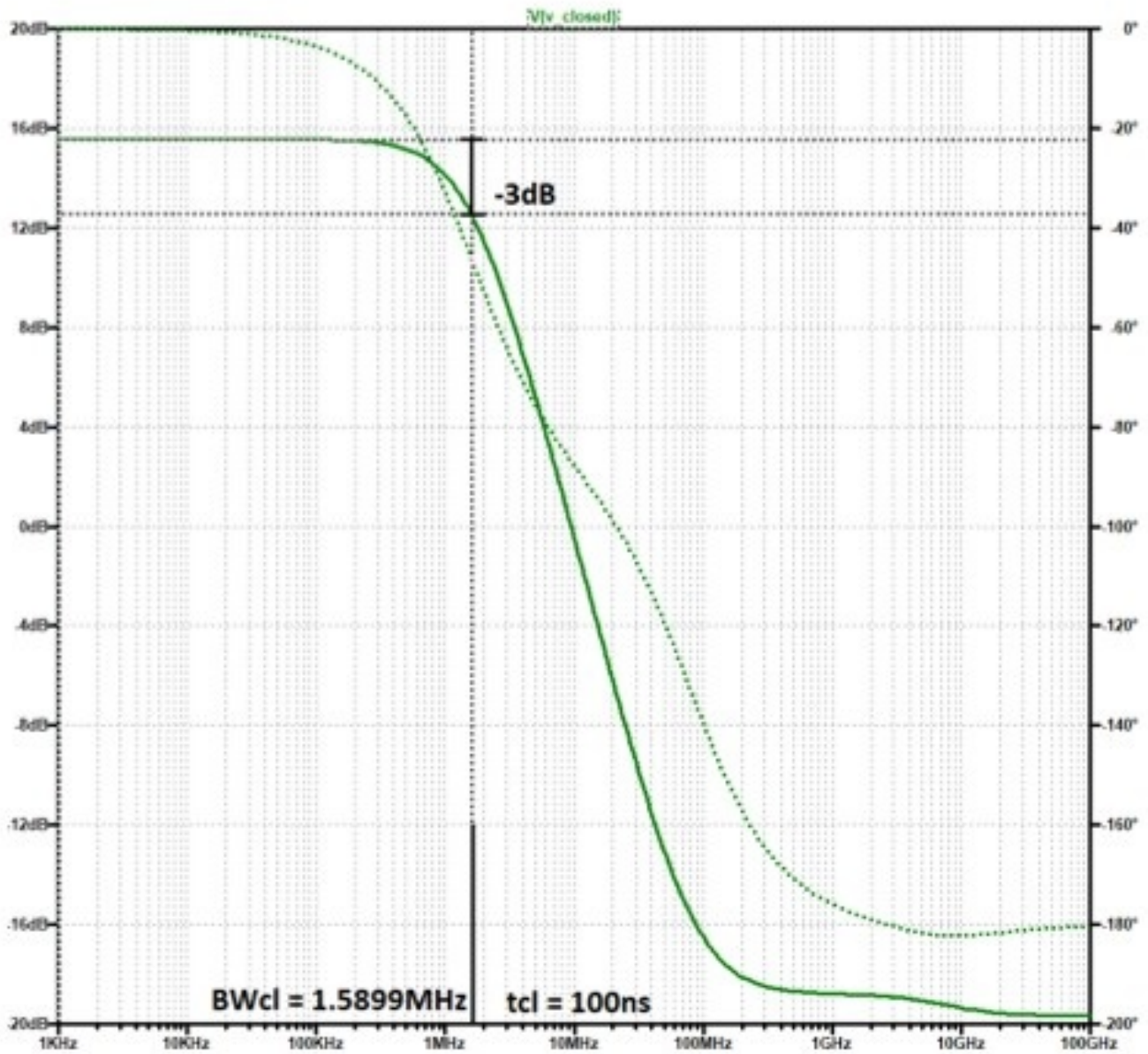


Figure 4: AC response of Closed-loop

Settling accuracy versus time

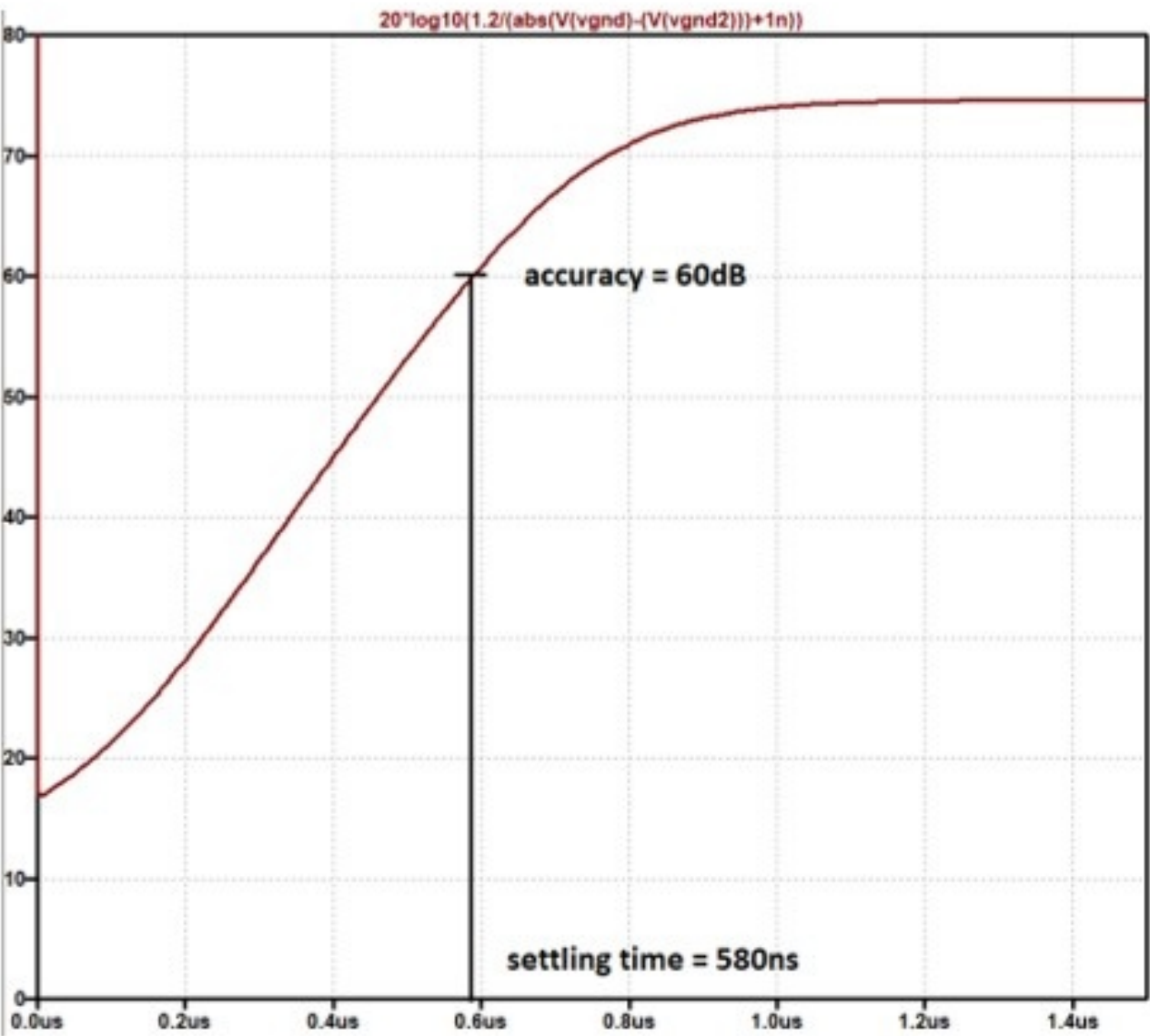


Figure 5: Settling behaviour

Settling behavior with T_{30dB} and T_{50dB} indicated

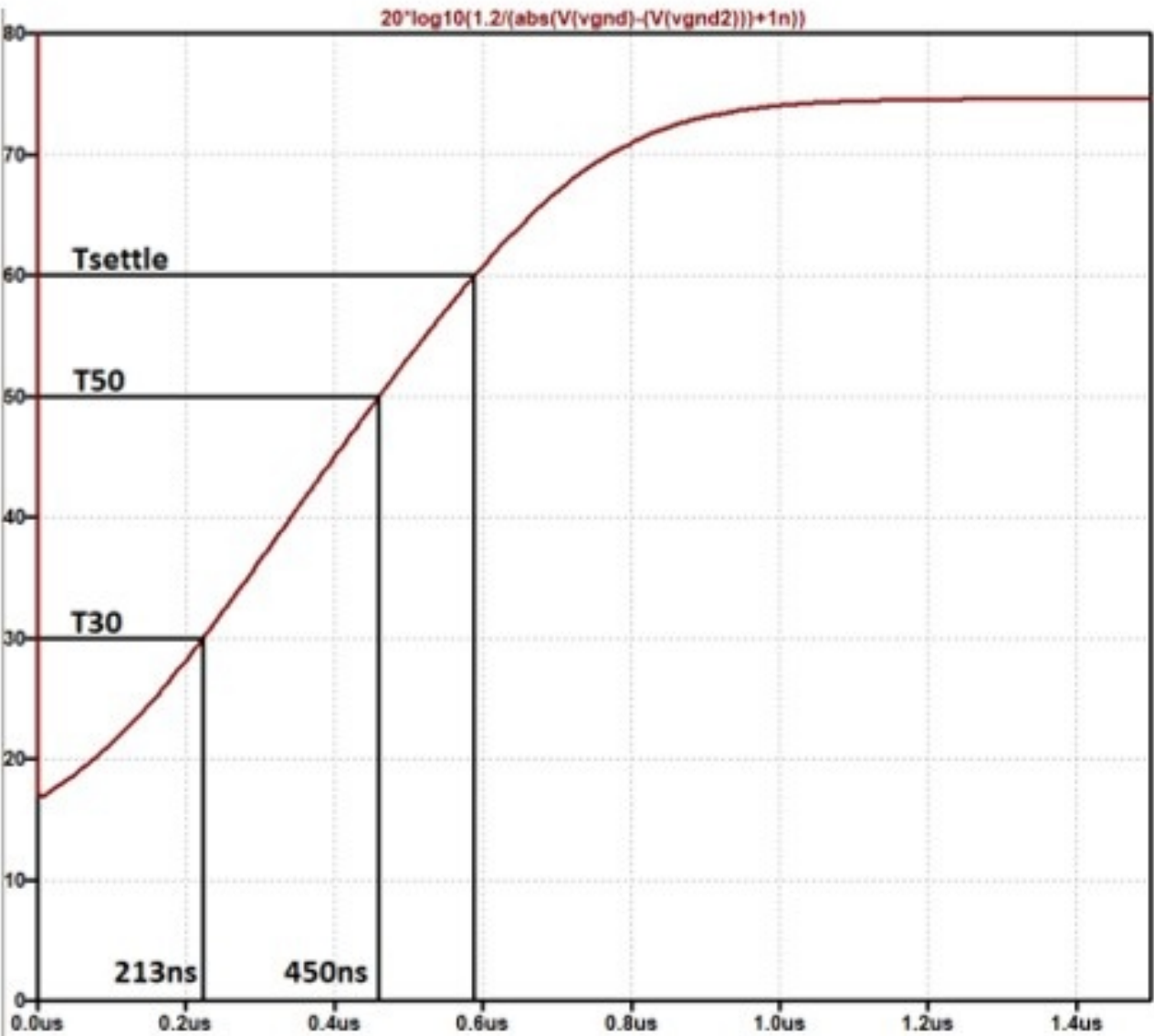


Figure 6: Settling behavior with times and magnitudes indicated

Output noise power density versus frequency

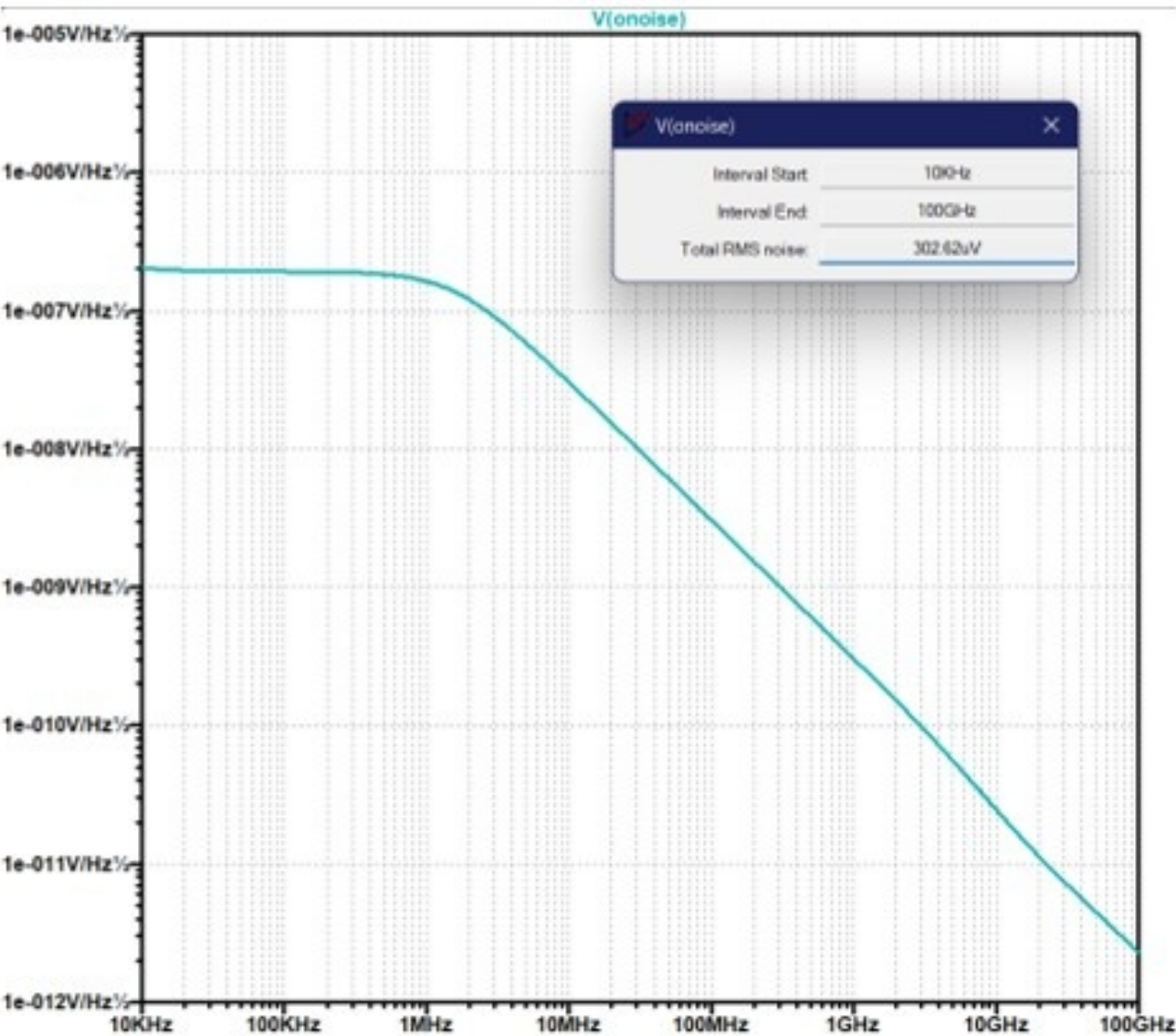


Figure 7: Output noise; the floating window displays the total integrated noise

Short Explanation

Observations

- Increasing the scaleA factor improves the SNR, speeds up settling, but requires more power for the same reference current.
- The key element affecting the SNR are the capacitor values. Larger capacitors reduce the output noise but also slow down the amplifier's settling speed.
- To force input pair transistors into weak inversion and hence increase gm-efficiency, they must be wide (more gm for the same bias current). Furthermore, they must be noise-optimized.
- To reduce the impact of thermal noise from current source transistors, strong inversion biasing is required. To achieve this, a greater current must flow through them.
- The transistor input pair size is also influencing the settling time, affecting the load of the input signal.

Brief explanation of design process

1. Finding the appropriate numbers for Cin and scaleA in order to adhere to the requirements was the first stage in the design process. In particular, the scaleA and the capacitor factor have been progressively reduced while aiming to reach the desired settling time.
2. Afterwards, the input pair size is increased, affecting them the settling time and the total integrated noise; therefore an optimal size have been reached.
3. Eventually, the specifications are met by the design by modifying both the biasing current and the input capacitor, keeping scaleA and input pair fixed.

To achieve the minimal power consumption the design process is iterated in its last phase every time the scaleA factor was reduced.

Conclusions

- With the aid of the open loop analysis, it is possible to see the curves A and $A\beta$ against frequency; the magnitude of the difference represents the inverse of closed-loop in band gain. Since the actual gain is given by $\frac{A}{1+A\beta}$, which becomes nearly $1/\beta$ for very high A: the higher the open loop gain, the more accurate the Op-amp gain should be.
- The cut-off frequency is reached when the amplifier's closed loop gain drops by 3 dB, or when the $A\beta$ in open loop simulations crosses 0 dB.
- System stability is ensured by the phase and gain margins.